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Exam No:

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KINETICS AND ESTER HYDROLYSIS

Aim:

To determine the rate constant of the first order ester hydrolysis reaction.

Apparatus Required:

i, Conical flasks (250ml)
ii, Burette (50ml)
iii, Pipette
iv, Ice box and ice cold water
v, Stopwatch
vi, Thermostat
vii, Measuring Cylinder (100ml, 25ml, 5ml)

Chemicals Required:

i, Hydrochloric Acid (0.4N)
ii, Sodium Hydroxide Solution (0.1N)
iii, Pure ethyl acetate
iv, Phenolphthalein indicator

Theory:

The hydrolysis of ethyl acetate ester catalyzed by an acid (HCl) may be represented as

$$\underset{\text{(ester)}}{CH_3COOC_2H_5} + \underset{\text{(water)}}{H_2O} + H^+ \longrightarrow \underset{\text{(acid)}}{CH_3COOH} + \underset{\text{(alcohol)}}{C_2H_5OH} + H^+$$

In this reaction water is present in large excess and the change in its concentration during the course of

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reaction is negligible and the concentration of acid remains constant throughout the reaction, the reaction is of first order. The rate of hydrolysis can be determined by taking out a definite volume of reaction mixture from time to time, and at the end of the hydrolysis and titrating it with alkali. The expression for rate constant k for the first order reaction at a particular temperature is given by

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

(or)

$$k = \frac{2.303}{t} \log \frac{V_\infty - V_0}{V_\infty - V_t}$$

[in terms of volume of alkali assuming the reaction goes to completion]

Here V_0, the volume of alkali required at zero minute, which is very difficult to obtain, as hydrolysis starts as soon as the ester is mixed with HCl solution and it can be eliminated by taking the difference Δt_n ($\Delta t_n = t_n - t_1$) such that

$$\Delta t_n = t_n - t_1 = \frac{2.303}{k} \log \frac{V_\infty - V_1}{V_\infty - V_n} \quad (n = 2, 3, 4 \ldots \text{etc})$$

where,

V_1 = volume of alkali at required time t_1

($V_1 \propto$ amount of HCl present originally + amount of acetic acid produced at time t_1)

V_n = volume of alkali required at time t_n

V_∞ = volume of alkali required when the hydrolysis of ester is complete.

Observation: (Table)

S.No	t_n(min)	V_n(ml) (volume of alkali used)
1.	$5 - t_1$	$V_1 = 19.0$ ml
2.	$10 - t_2$	$V_2 = 20.0$ ml
3.	$20 - t_3$	$V_3 = 21.8$ ml
4	$30 - t_4$	$V_4 = 22.5$ ml
5	$50 - t_5$	$V_5 = 25.0$ ml
6.	$100 - t_\infty$	$V_\infty = 51.6$ ml

Room temperature : 25°C

SNO	$\Delta t_n = t_n - t_1$ (min)	$V_\infty - V_n$ (ml)	$\log\left(\frac{V_\infty - V_1}{V_\infty - V_n}\right)$
1.	$t_1 - t_1 = 0$	$V_\infty - V_1 = 32.6$ ml	0
2.	$t_2 - t_1 = 5$ min	$V_\infty - V_2 = 31.6$ ml	0.0135
3.	$t_3 - t_1 = 15$ min	$V_\infty - V_3 = 29.8$ ml	0.0390
4.	$t_4 - t_1 = 25$ min	$V_\infty - V_4 = 29.1$ ml	0.0493
5.	$t_5 - t_1 = 45$ min	$V_\infty - V_5 = 26.6$ ml	0.0883

Plot of $\log (V_\infty - V_1 / V_\infty - V_n)$ vs Δt_n gives a straight line passing through origin and from the slope of which the value of k_s can be calculated.

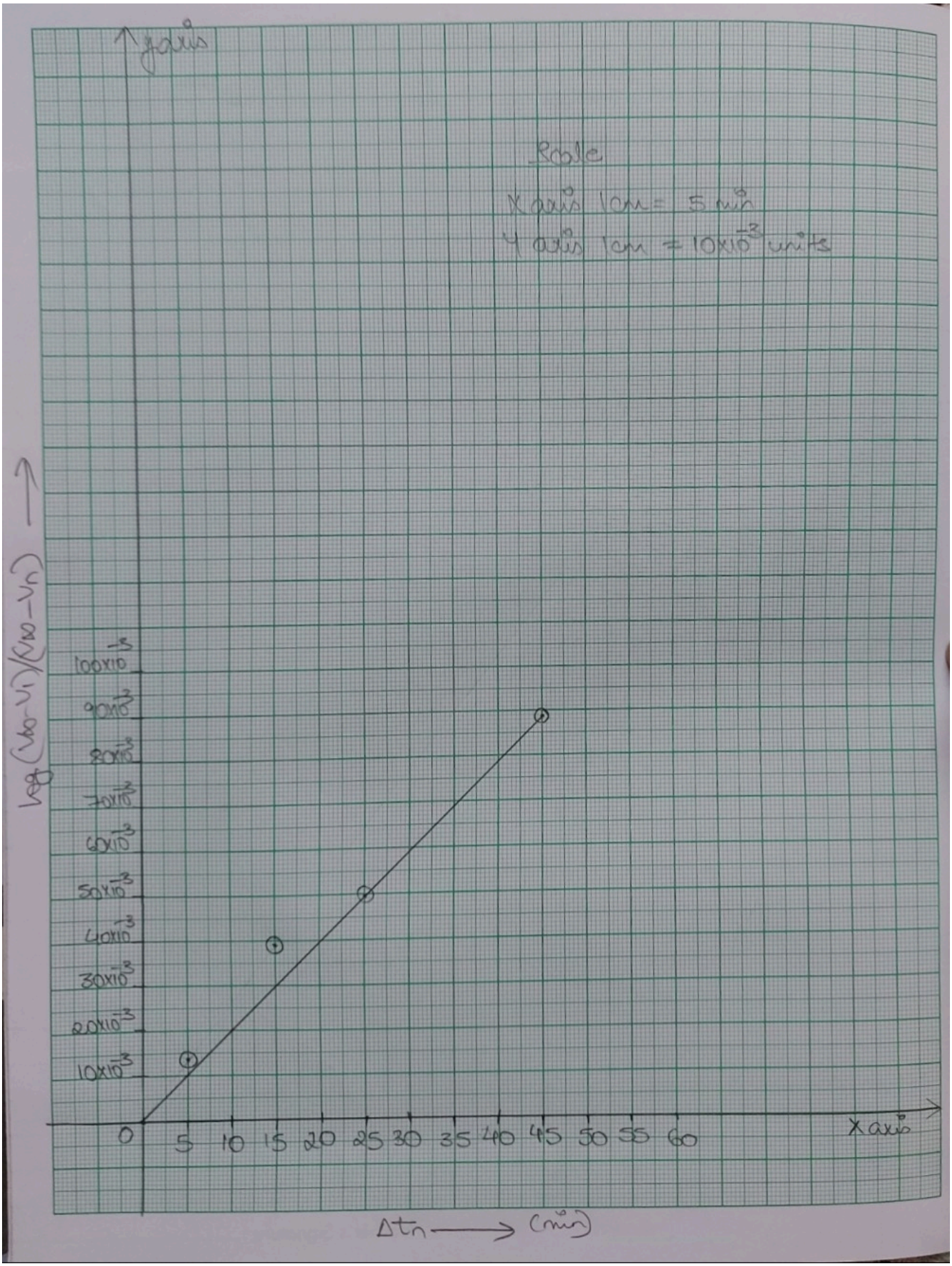
Procedure:

i) 100ml (0.4N) HCl solution was taken by a measuring cylinder in a dry conical flasks. Then 10ml ethyl acetate was pipetted out and poured into 100ml (0.4N) HCl solution kept in the conical flasks. The stopwatch was started when the pipette was half emptied. The conical flasks was covered and the reaction mixture was shaken. [The stop watch was not stopped until the experiment ended]

ii) After 5 minutes, 5ml of the reaction mixture was taken out and poured it into 100ml conical flasks which had 25ml ice cold water, then it was titrated with NaOH solution of strength 0.1N using phenolphthalein indicator as quick as possible. This step was repeated at 10, 20, 30, 50 and 80 minutes

iii) In the meantime, 20ml of the reaction mixture was pipetted out and poured into a conical flasks. Then it was placed in water bath kept at 50°C - 60°C temperature for a period of 100 minutes to allow the ester to get completely hydrolyzed. After this period, it was taken out and allowed to cool to room temperature. 5ml of this mixture was pipetted out into a 100ml conical flasks and titrated like before we have done.

iv) The $\log \left(\frac{V_\infty - V_1}{V_\infty - V_n}\right)$ vs Δt_n was plotted. This generated


y axis
Scale
X axis 1cm = 5 min
Y axis 1cm = 10x10^-3 units
log (V∞-V1)/(V∞-Vn)
100x10^-3
90x10^-3
80x10^-3
70x10^-3
60x10^-3
50x10^-3
40x10^-3
30x10^-3
20x10^-3
10x10^-3
0
5 10 15 20 25 30 35 40 45 50 55 60
X axis
Δtn → (min)

Calculations:

From graph,

$$\text{slope} = \frac{0.0493 - 0}{25 - 0}$$

$$= 1.972 \times 10^{-3}$$

$$\Rightarrow \frac{K}{2.303} = 1.972 \times 10^{-3}$$

$$K = 2.303 \times 1.972 \times 10^{-3} \text{ min}^{-1}$$

$$= 4.54 \times 10^{-3} \text{ min}^{-1}$$

Rate constant (K) = $4.54 \times 10^{-3} \text{ min}^{-1}$

a straight line graph passing through origin with slope $= \frac{K}{2.303}$. Therefore K was calculated from the slope of the graph. (K = rate constant)

Result :

The hydrolysis of ethyl acetate is an example of first order reaction with rate constant (K) $= 4.54 \times 10^{-3}\ min^{-1}$

Precautions :

i, The titrations are measured with utmost care and values are noted as soon as solution turns light pink.
ii, The concentrated solutions are handled with safety.
iii, Accurate measurements should be taken.

Conclusion :

In this experiment, we saw that the graph of $\log (V_\infty - V_i)/(V_\infty - V_n)$ vs Δt_n is a line with a slope of $K/2.303$. The hydrolysis of ethyl acetate is an example of first order reaction. Using this we found that the rate of the reaction to be $4.54 \times 10^{-3}\ min^{-1}$.

Discussion :

The rate of reaction is often proportional to the concentration of the reactants to some power. The proportionality constant is called rate constant. This rate constant doesnot depend upon the concentration of the reactant but it does on the temperature.

So, the readings can fluctuate due to the temperature. The order of reaction is nothing but the power dependence of the rate of reaction on the concentration of the all reactants. The reaction order is always defined with the help of reactant concentration, not the product concentration.